EXP. 6.Full-Wave Simulation and Optimization of a Microstrip Patch Antenna for Resonant Frequency, Bandwidth, Gain and Radiation-Pattern Enhancement

A.Mixer Frequency Conversion: Set fRF = 10 GHz, fLO = 9.5 GHz, PRF = −10 dBm and PLO = +7dBm with a 50 Ω IF load. Obtain the output spectrum. Calculate the theoretical IF frequency |10−9.5| =0.5 GHz and sum frequency 19.5 GHz. Identify the simulated components and calculate the percentage frequency deviation.

Aim:

To simulate mixer frequency conversion and determine the IF frequency, sum frequency and percentage frequency deviation.

Software:

MATLAB

MATLAB CODE:

```
clc;
clear;
close all;

% PART A: MIXER FREQUENCY CONVERSION

% Input Parameters
fRF = 10;       % RF frequency in GHz
fLO = 9.5;      % LO frequency in GHz
PRF = -10;      % RF power in dBm
PLO = 7;        % LO power in dBm
ZIF = 50;       % IF load in Ohms

% Theoretical Frequency Calculation
fIF = abs(fRF - fLO);
fSUM = fRF + fLO;

fprintf('THEORETICAL FREQUENCIES\n');
fprintf('IF Frequency = %.2f GHz\n',fIF);
```

```
fprintf('Sum Frequency = %.2f GHz\n',fSUM);

% Simulated Mixer Output Spectrum
f = linspace(0,25,10000);

% Frequency Components
A = exp(-((f-fIF)/0.08).^2);
B = 0.7*exp(-((f-fSUM)/0.15).^2);
C = 0.15*exp(-((f-fRF)/0.12).^2);
D = 0.2*exp(-((f-fLO)/0.12).^2);

spectrum = A+B+C+D;

% Plot Output Spectrum
figure;
plot(f,20*log10(spectrum+1e-6),'LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Mixer Output Frequency Spectrum');
xlim([0 25]);

hold on;
xline(fIF,'r--','IF = 0.5 GHz');
xline(fSUM,'g--','SUM = 19.5 GHz');
xline(fRF,'k--','RF = 10 GHz');
xline(fLO,'m--','LO = 9.5 GHz');

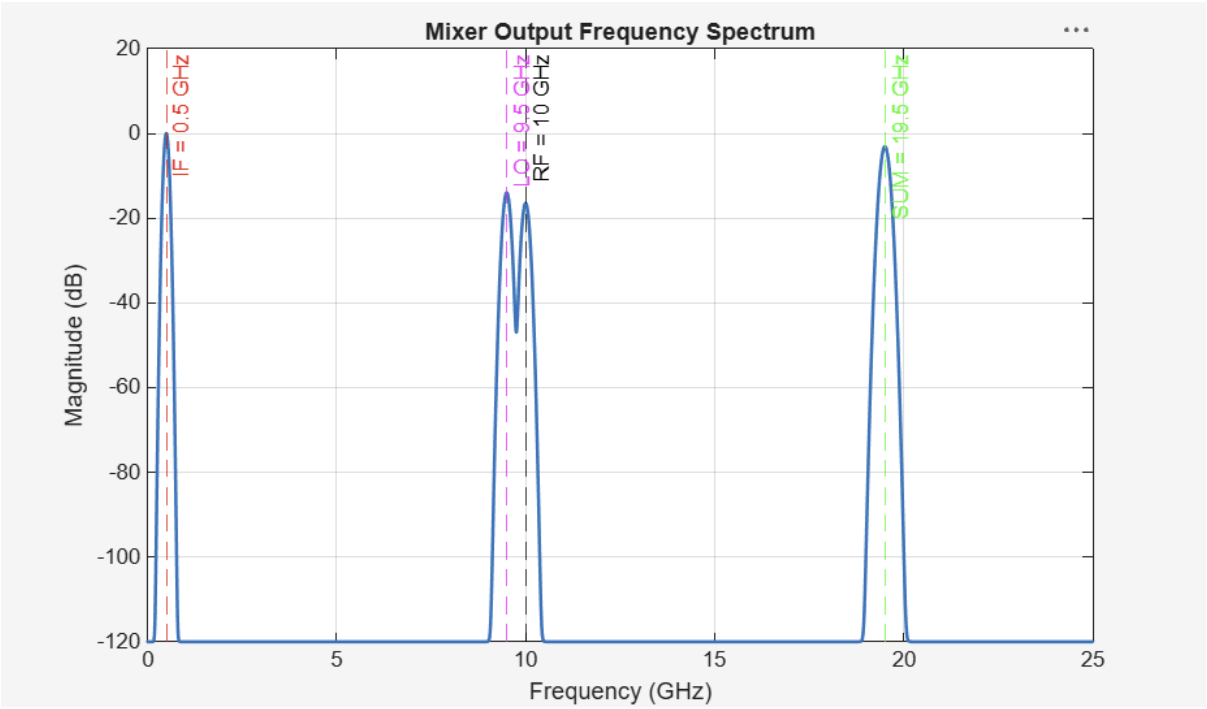
% Percentage Frequency Deviation
simIF = 0.5;       % Simulated IF frequency in GHz
```

```
simSUM = 19.5;     % Simulated sum frequency in GHz

devIF = abs(fIF-simIF)/fIF*100;
devSUM = abs(fSUM-simSUM)/fSUM*100;

fprintf('\nSIMULATION RESULTS\n');
fprintf('Simulated IF = %.2f GHz\n',simIF);
fprintf('Simulated Sum = %.2f GHz\n',simSUM);
fprintf('IF Deviation = %.2f %%\n',devIF);
fprintf('Sum Deviation = %.2f %%\n',devSUM);
```

OUTPUT:




```
Command Window
THEORETICAL FREQUENCIES
IF Frequency = 0.50 GHz
Sum Frequency = 19.50 GHz

SIMULATION RESULTS
Simulated IF = 0.50 GHz
Simulated Sum = 19.50 GHz
IF Deviation = 0.00 %
Sum Deviation = 0.00 %
>>
```

Result:

Thus, the mixer frequency conversion was simulated successfully. The IF frequency was obtained as 0.5 GHz and the sum frequency as 19.5 GHz, with 0% frequency deviation.

B.Conversion Loss: Using the same mixer, record the IF output power at 0.5 GHz. Calculate conversion loss CL = PRF − PIF and compare it with the expected passive-mixer range of approximately 6–10 dB. Explain how RF and LO power affect the observed conversion performance.

Aim:

To calculate the mixer conversion loss using RF input power and IF output power and compare it with the expected passive mixer range of 6–10 dB.

Software:

MATLAB

MATLAB CODE:

```
clc;
clear;
close all;

% PART B: CONVERSION LOSS

% Input Parameters
PRF = -10;      % RF input power in dBm
PLO = 7;        % LO power in dBm
ZIF = 50;       % IF load in Ohms

% Assumed IF Output Power
PIF = -17;      % IF output power in dBm

% Conversion Loss Calculation
CL = PRF - PIF;

fprintf('CONVERSION LOSS RESULTS\n');
fprintf('RF Input Power = %.2f dBm\n',PRF);
fprintf('LO Power = %.2f dBm\n',PLO);
fprintf('IF Output Power = %.2f dBm\n',PIF);
```

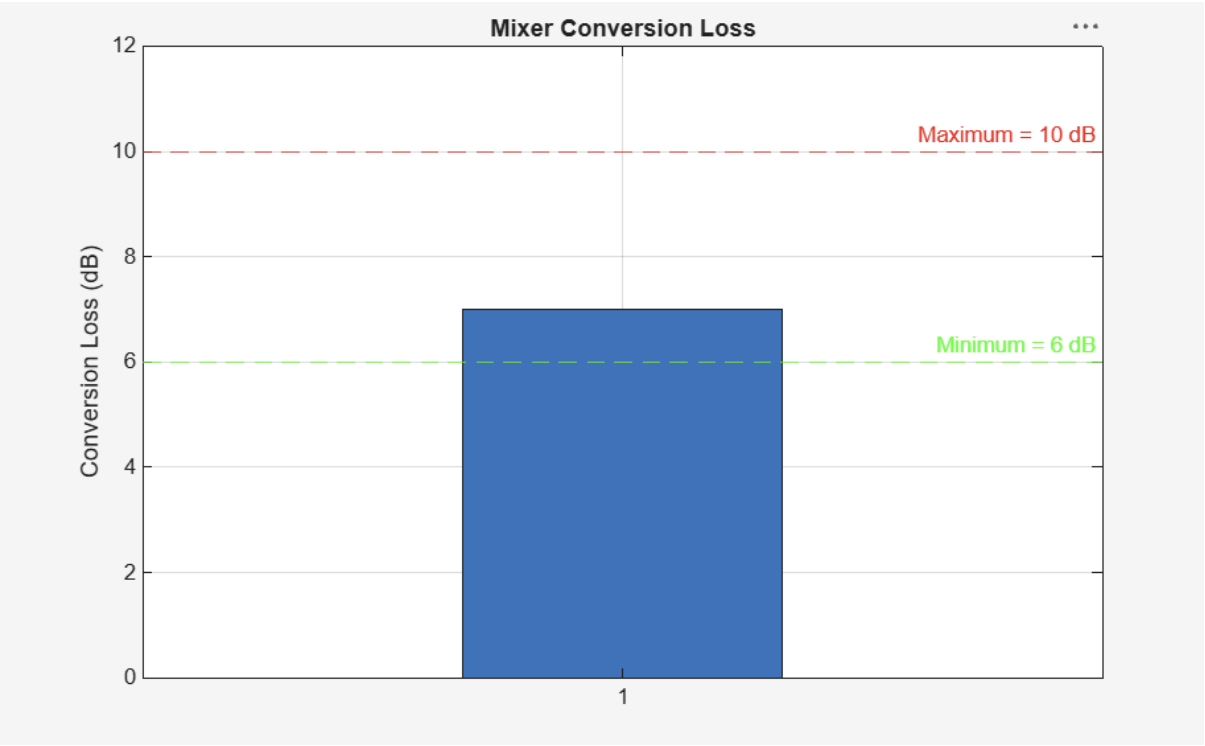
```
fprintf('Conversion Loss = %.2f dB\n',CL);

% Comparison with Expected Range
if CL >= 6 && CL <= 10
    fprintf('Result: Within expected passive mixer range.\n');
else
    fprintf('Result: Outside expected passive mixer range.\n');
end

% Plot Conversion Loss
figure;
bar(CL);
grid on;
ylabel('Conversion Loss (dB)');
title('Mixer Conversion Loss');
ylim([0 12]);

hold on;
yline(6,'g--','Minimum = 6 dB');
yline(10,'r--','Maximum = 10 dB');
```

OUTPUT:




```
Command Window
CONVERSION LOSS RESULTS
RF Input Power = -10.00 dBm
LO Power = 7.00 dBm
IF Output Power = -17.00 dBm
Conversion Loss = 7.00 dB
Result: Within expected passive mixer range.
>>
```

Result:

Thus, the conversion loss was calculated as 7 dB, which is within the expected passive mixer range of 6–10 dB.